INCH-POUND

MIL-PRF-44156E (GL)
1 March 2006
SUPERSEDING
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24 February 1997

PERFORMANCE SPECIFICATION

KITCHEN, COMPANY LEVEL, FIELD FEEDING (KCLFF), COMPONENTS THEREOF

This specification is approved for use by the Natick Research, Development, and Engineering Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the fabricated components of the Kitchen, Company Level, Field Feeding (KCLFF).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in section 3 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in section 3 of this specification, whether or not they are listed.

Comments, suggestions or questions on this document should be addressed to US Army Research, Development and Engineering Command, Natick Soldier Center, AMSRD-NSC-CF-S, 15 Kansas St., Natick, MA 01760-5018 or emailed to Robert.Bernazzani@natick.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil

AMSC N/A FSC 7360

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2.2 Government documents. (None).

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

NSF INTERNATIONAL

STANDARD NO. 2 - Food Service Equipment
STANDARD NO. 4 - Commercial Cooking and Hot Food Storage Equipment
CRITERIA C-2 - Special Equipment and/or Devices

(Application for copies should be addressed to NSF International, 789 N. Dixboro Road P.O. Box 130140, Ann Arbor, MI 48113-0140).

2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), KCLFF shall be subjected to first article inspection in accordance with 4.1.1.

3.2 **Heater Tank operating requirements**. The KCLFF component identified in Table I (reference figure 1) shall satisfy the following requirements.

TABLE I. Heater Tank component for KCLFF

Item No.	Description	Reference Paragraph	Quantity	Maximum Weight (lbs)
1	Heater tank assembly, tray pack	3.2.1 thru 3.2.6	1	88

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3.2.1 Heater tank, tray pack, size and configuration. The tray pack heater tank (reference figure 1) shall have a capacity of not less than 20 gallons of water and 20 tray packs (reference figure 2) and shall have a visual mark on the inside at the 20-gallon level. The tank assembly shall be approximately 37 inches high and shall accommodate a Modern Burner Unit(MBU) (see 6.6.1). The tank assembly shall have two handles on the front and two on the rear that fold down along the tank's vertical surfaces when not in use. When the handles are utilized, they shall not rotate upward more than approximately 90 degrees. The heater tank shall be provided with a two- section cover, split in the center, hinged on each side and capable of being secured in the closed position with latches. Each tank shall contain a drain, a drain ball valve and a drain pipe with a hose adapter.

3.2.2 Heater tank, tray pack, heating. The heater tank assembly shall permit water to be rapidly heated, and shall withstand direct heat from the burner unit (see 6.6.1).

3.2.3 Heater tank, tray pack, drainage. The tank drain shall permit rapid drainage.

3.2.4 Heater tank, tray pack, leakage. The tank shall not leak.

3.2.5 Heater tank, tray pack, warning plates. Warning plates that specify, "Warning, May Be Hot" shall be mounted to the front and back of the tank. Marking shall be durable.

3.2.6 Heater tank, tray pack, operating instructions plate. An operating instruction plate shall be provided on the front of the tank assembly and shall specify the following:

"1. Place burner in rack
2. Fill with 20 gallons of water for full load of 20 tray packs
3. Light burner per TM 10-7310-281-13 &P
4. Bring water to a boil leaving burner on high setting
5. Place tray packs in tank being sure they are completely covered with water
6. Heat tray packs for 45 minutes
8. Remove tray packs using tray pack lifter
9. Shut burner off by pressing stop"

Marking shall be durable.

3.3 **Griddle operating requirements**. The KCLFF component identified in table II (reference figure 3) shall satisfy the following requirements.

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TABLE II. Griddle and base for KCLFF

Item No.	Description	Reference Paragraph	Quantity	Maximum Weight (lbs)
1	Griddle	3.3.1, 3.3.2 and 3.3.3	1	14
2	Griddle base	3.3.4	1	8

3.3.1 Griddle, size and configuration. The griddle shall be approximately 21 inches long, 18 inches wide and 1-5/8 inches high, excluding handles, in order to interface with each griddle base. Two handles shall be provided that shall have openings no smaller than 4 inches wide and 1-1/8 inches high.

3.3.2 Griddle, flatness. The griddle shall remain flat after repeated heating and cooling.

3.3.3 Griddle, surface. The griddle shall have a smooth surface that is nontoxic and shall not impart odors, color or taste, or otherwise contribute to the adulteration or contamination of food during normal operating temperatures.

3.3.4 Griddle base, size and configuration. The griddle shall fit within a rim in the griddle base. The griddle base shall interface with the burner rack and base (see 6.6.1) that are utilized with the cook pot cradle assembly (see 6.7.1).

3.3.5 Chemical resistance. Components listed in table I and table II shall be resistant to the detergents and sanitizers used for washing and sanitizing operations.

3.3.6 Safety. Components listed in table I and table II shall be safe to set up, operate, and transport and shall not create personnel hazards.

3.3.7 Sanitation/Cleaning. Food service equipment shall meet the sanitation requirements of NSF/ANSI Standard 2, Food service equipment. Surfaces shall be smooth, washable, free of unnecessary ledges, projections or crevices, and readily accessible for cleaning.

3.4 **Integrated equipment.** The KCLFF shall also include the components listed under section 3.4.1.

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3.4.1 KCLFF. Each complete KCLFF shall include all of the equipment listed below.

Item No.	Description	National Stock Number	Quantity
1	Adapter, electrical	7310-01-454-1241	1
2	Adapter, fuel	7310-01-455-3736	1
3	Bag, drinking water storage	4610-00-268-9890	1
4	Batteries	7310-01-454-1249	1
5	Battery Pack	7310-01-453-6565	1
6	Board, food chopping	7330-00-078-5706	1
7	Modern Burner Unit, (MBU)	7310-01-507-9310	3
8	Cable 4, DC extension	7310-01-455-0665	1
9	Cable Clip	7310-01-457-5836	1
10	Cable 3, 2-branch	7310-01-455-1017	1
11	Cable 6	7310-01-509-4453	1
12	Can, fuel, military: plastic, 5-gallon	7240-01-337-5269	2
13	Can opener, heavy duty	7330-01-411-9789	1
14	Can, water, plastic, 5-gallon	7240-00-089-3827	8
15	Chest, ice storage	4110-01-452-7317	1
16	Cook pot cradle assembly	7330-01-248-9964	1
17	Cover, cooking pot	7330-00-250-6300	1
18	Dispenser, liquid, insulated, 5-gallon	7310-01-245-6937	4
19	Extinguisher, fire	4210-00-270-4512	1
20	Field range outfit, and field range M59A	7360-01-479-0312	1
21	First aid kit, general purpose	6545-00-919-6650	1

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22	Food container, insulated with inserts	7360-01-408-4911	16
23	Fuel line, 20 feet	7310-01-455-3735	1
24	Gloves, heat protective	8415-01-092-3910	1 pair
25	Griddle assembly, field	7310-01-391-3065	1
26	Heater tank assembly	7360-01-248-6041	1
27	Hose	4720-00-729-5334	1
28	Knife, boning	7340-00-197-1271	1
29	Knife, slicing	7340-00-406-6531	1
30	Lantern, diesel	6260-01-535-1647	1
31	Lifter assembly, tray pack, extracting	7330-01-234-2164	1
32	Lifter assembly, tray pack, serving	7330-01-224-0914	1
33	Lifter serving, number ten can	7310-01-385-6263	1
34	Measure, liquid, 2-quart	7330-00-205-3096	1
35	Nut driver	5120-01-335-0349	1
36	Opener, can, hand, for tray packs and round cans	7330-01-245-0201	2
37	Pot, cooking (without cover), 10-gal.	7330-00-292-2306	1
38	Pot, cooking (with cover), 15-gal.	7330-00-292-2307	1
39	Screwdriver, cross tip	5120-00-234-8913	1
40	Screwdriver, flat tip	5120-00-222-8852	1
41	Spoon, food service, basting, 21-inch	7340-00-223-7800	4
42	Spoon, food service, slotted, 15-inch	7340-00-205-1421	4
43	Spout, can flexible	7240-00-177-6154	1
44	Table, field	7105-01-500-0012	2
45	Wrench, adjustable	5120-00-240-5328	1

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3.5 **Interface and interoperability requirements**.

3.5.1 Interoperability and interchangeability. The components listed in table I and table II shall be functionally interoperable and physically interchangeable with components of the currently fielded KCLFF and KCLFF-E.

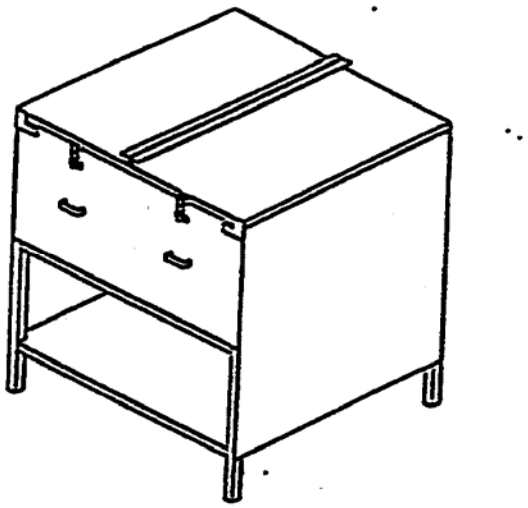


FIGURE 1. Tray Pack Heater Tank Assembly

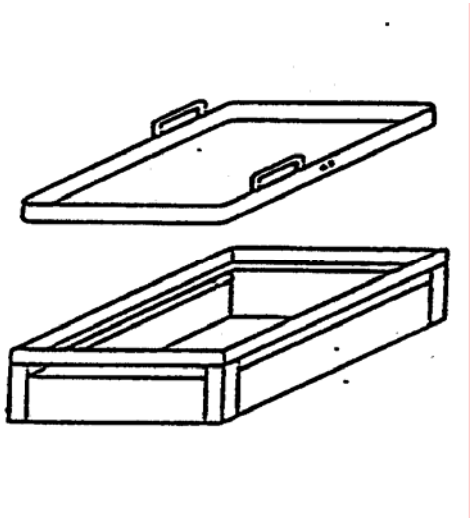


FIGURE 3. Griddle Assembly

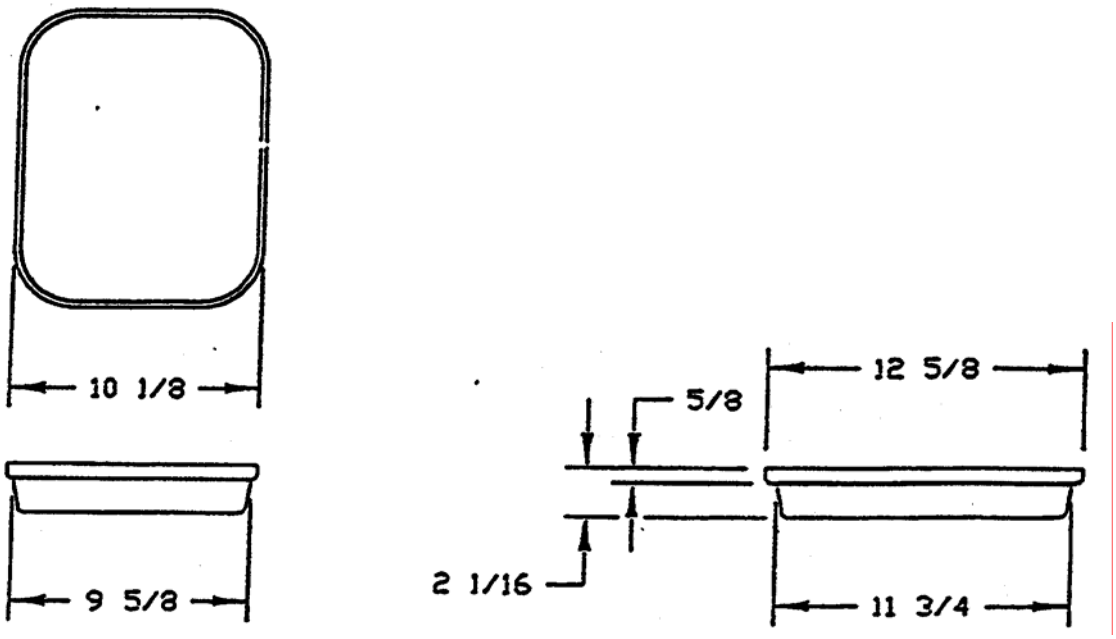


FIGURE 2. Tray Pack Can
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3.6 **Support or ownership requirements.**

3.6.1 Shelf life. The depot storage life shall be not less than 10 years.

3.6.2 Service life. All components shall have a service life of not less than 5 years of continuous use in a field environment.

3.6.3 Durability. Repeated operational use shall not cause the components to leak, fracture, deform or undergo dimensional changes that prevent acceptable performance or prevent interfacing with associated assemblies/components.

3.6.4 Weight. The weight of the KCLFF components shall not exceed the weights specified in table I and table II.

3.7 **Environmental requirements.** The components listed in table I and table II shall operate and be functional in ambient temperatures from -25^{0}F to $+125^{0}$F.

3.7.1 Corrosion resistance. Components listed in table I and table II shall be corrosion resistant.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as first article inspection and conformance inspection.

4.1.1 First article inspection. When a first article inspection is required (see 3.1 and 6.2) the first article shall be examined in accordance with table III. Sampling for first article inspection shall be as specified in the contract or purchase order.

4.1.2 Conformance inspection. Conformance inspection includes those examinations and tests from table III, as defined in the contract or purchase order, performed on specified samples (see 6.2 and 6.4).

4.2 Order of verification. Environmental tests shall be conducted prior to remaining verifications which may be conducted in any sequence.

4.3 Verification methods. The types of verification methods may be visual inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously-approved or previously-qualified designs.

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4.3.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost-effective sampling procedures to verify performance. The contract may specify alternatives that replace verifications required by this specification.

TABLE III. Matrix - Requirements vs verification methods

Title	Requirement	Verification
HEATER TANK OPERATING REQUIREMENTS	3.2	4.3.2
Heater tank, tray pack, size and configuration	3.2.1	4.3.2.1
Heater tank, tray pack, heating	3.2.2	4.3.2.2
Heater tank, tray pack, drainage	3.2.3	4.3.2.3
Heater tank, tray pack, leakage	3.2.4	4.3.2.4
Heater tank, tray pack, warning plates	3.2.5	4.3.2.5
Heater tank, tray pack, operating instructions plate	3.2.6	4.3.2.6
Griddle, size and configuration	3.3.1	4.3.2.7
Griddle, flatness	3.3.2	4.3.2.8
Griddle, surface	3.3.3	4.3.2.9
Griddle base, size and configuration	3.3.4	4.3.2.10
Chemical resistance	3.3.5	4.3.2.11
Safety	3.3.6	4.3.2.12
Sanitation/Cleaning	3.3.7	4.3.2.13
INTERFACE AND INTEROPERABILITY REQUIREMENTS	3.5	4.3.3
Interoperability and interchangeability	3.5.1	4.3.3.1
SUPPORT OR OWNERSHIP REQUIREMENTS	3.6	4.3.4
Shelf life	3.6.1	4.3.4.1
Service life	3.6.2	4.3.4.2
Durability	3.6.3	4.3.4.3
Weight	3.6.4	4.3.4.4

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ENVIRONMENTAL REQUIREMENTS	3.7	4.3.5
Corrosion resistance	3.7.1	4.3.5.1

4.3.2 **Operating requirements verification.**

4.3.2.1 Heater tank, tray pack, size and configuration inspection. Inspect the heater tank and verify that there is a visual mark on the inside and demonstrate that the visual mark is at the 20-gallon level, and that the addition of 20 tray packs (reference figure 2) is within the capacity of the tank. Measure and verify the height of the tank assembly. Demonstrate that a Modern Burner Unit(MBU) will fit into the heater tank assembly. Inspect the tank assembly and verify that there are two fold-down handles on each end with upward motion limited to 90 degrees and covers on the top, hinged on each side, with latches to secure the covers in the closed position. Inspect the tank assembly and verify that there is a drain, a drain ball valve and a drain pipe with a hose adapter.

4.3.2.2 Heater tank, tray pack, heating demonstration. Demonstrate tank heating capability by filling the tank with 20 gallons of water. The temperature of the water shall be 60^0F $\pm$ 5^0F. The ambient temperature during testing shall be 70^0F $\pm$ 10^0F and wind velocity shall be no more than 2 mph. Verify that the tank water is heated to 212^0F in not more than 1 hour. Verify that the tank does not have damage caused by burner heat, such as distortion or pitting.

4.3.2.3 Heater tank, tray pack, drainage demonstration. Demonstrate the drainage capability of the heater tank by filling the tank with 20 gallons of water, and opening the drain valve completely. Verify that the water drains from the tank to a level less than 3/4 inch deep in not more than 6 minutes.

4.3.2.4 Heater, tank, tray pack, leakage inspection. Visually inspect the heater tank during and subsequent to the heating and drainage demonstrations. Verify that leakage has not occurred.

4.3.2.5 Heater tank, tray pack, warning plates inspection. Inspect the warning plates. Verify the warning plate text and location.

4.3.2.6 Heater tank, tray pack, operating instructions plate inspection. Inspect the operating instruction plate. Verify the text and location.

4.3.2.7 Griddle, size and configuration inspection. Measure and verify the length, width, and height of the griddle top and the width and height of the handle openings.

4.3.2.8 Griddle, flatness demonstration. Verify that the griddle cooking surface flatness is within 1/8 inch, when measured from a straightedge placed across each set of diagonal corners, after six cycles of heating to not less than 375^0F and cooling to room ambient temperature.

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4.3.2.9 Griddle, surface verification. The contractor shall verify that the griddle's surface complies with the requirements of NSF International Standards Number 4 and Criteria C-2.

4.3.2.10 Griddle base, size and configuration inspection. Demonstrate that the griddle will fit within the rim of the griddle base. Demonstrate that the base will interface with a burner rack (see 6.6.1).

4.3.2.11 Chemical resistance verification.

4.3.2.11.1 Dishwashing compound. Dishwashing compound (NSN 7930-00-281-4731) shall be added to the water during the heating demonstration of the tray pack heater. The griddle shall be soaked for two hours in the dishwashing solution and rinsed in clear water. Upon completion of the demonstration, inspect the tray pack heater tank and griddle to verify that there is no evidence of chemical degradation.

4.3.2.11.2 Disinfectant. Heat 20 gallons of water in a tray pack heater tank to a temperature of 100^0F $\pm$ 10^0F. Dissolve one 4.77 ounce packet of food service disinfectant (NSN 6840-00-810-6396) in the water and allow the mixture to remain in the tank at 100^0F for 2 hours. Drain the tank and inspect the tray pack heater tank to verify that there is no evidence of chemical degradation.

4.3.2.12 Safety verification. Visually inspect the components and verify that they are free of hazardous burrs, nicks, sharp edges, foreign materials and other conditions that are hazardous to personnel during setup, operation and transport of the components.

4.3.2.13 Sanitation/Cleaning verification. Verify that food service equipment meet the requirements of NSF/ANSI Standard 2, Food service equipment. Verify that surfaces are smooth, washable, free of unnecessary ledges, projections or crevices, and are readily accessible for cleaning.

4.3.3 **Interface and interoperability requirements verification.**

4.3.3.1 Interoperability and interchangeability demonstration. Compare the test components in table I and table II with KCLFF components from current supply (see 6.6.2) following the procedures in 4.3.3.1.1 and 4.3.3.1.2.

4.3.3.1.1 Interface demonstration. Set up the test and the current KCLFF components. Verify the interchangeability of the test and current components.

4.3.3.1.2 Interoperability and interchangeability criteria. Visually examine each configuration to verify compatibility between the current components and the test items. Verify ease of operation. Verify that there are no incompatibilities affecting form, fit, function or total interchangeability.

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4.3.4 **Support or ownership requirements verification.**

4.3.4.1 Shelf life analysis. Perform an analysis to demonstrate that the materials and manufacturing processes of the KCLFF and KCLFF-E components listed in table I and table II are similar to those in the currently fielded KCLFF and KCLFF-E, or are capable of meeting the required shelf life.

4.3.4.2 Service life analysis. Perform an analysis to demonstrate that the materials and manufacturing processes of the KCLFF and KCLFF-E components listed in table I and table II are similar to those in the currently fielded KCLFF and KCLFF-E, or are capable of meeting the required service life.

4.3.4.3 Durability tests. The components listed in table I and table II shall undergo drop tests. Each component shall be dropped vertically from a height of 36 inches on two stacked 1/2-inch thick plywood sheets. Verify that no fractures, deformation or breakage occur that will prevent acceptable performance.

4.3.4.4 Weight demonstration. Weigh each of the components in table I and table II and verify that the maximum weight allowance specified in tables I and II is not exceeded.

4.3.5 **Environmental requirements verification**. Assemble the components listed in table I and table II after conditioning for 8 hours at $-25^{0}F \pm 5^{0}F$. Examine all components after assembly completion to verify that there is no evidence of failures.

4.3.5.1 Corrosion resistance. The components listed in table I and table II shall show no evidence of corrosion upon completion of the verifications contained in table III.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory).

6.1 Intended use. The company level field feeding kitchen is an assemblage of equipment that provides the capability of heating and serving tray pack rations (T-rations), A (semi-perishable and non-perishable) and B (semi-perishable) rations in basic, hot, and cold climatic conditions with or without shelter from the environment.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification
b. Type of kitchen required (see 1.2)
c. Issue of DoDISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.3)
d. When a first article is required (see 3.1, 6.3 and 6.6.1)
e. Sampling for conformance inspection (see 4.1.2)
f. Packaging requirements (see 5.1)
g. Source and delivery of government-loaned property (see 6.6)

6.3 First article. When requiring a first article inspection, the contracting officer should provide specific guidance to offerors whether the first article is a first article sample, a first production item, or a number of items to be tested. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Pre-solicitation documents should provide government waiver rights for samples for first article inspection to bidders offering a previously-acquired or tested product. Bidders offering such products, who wish to rely on such production or testing must furnish evidence with the bid, that prior government approval is appropriate for the pending contract.

6.4 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include contractor past performance, government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from table III and their designated frequency based on a risk assessment for the procurement.

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6.5 Technical data. KCLFF components listed in Table I and Table II have been produced satisfactorily using specification MIL-K-44156C (GL) and the drawings listed in the National Stock Number for the associated components under paragraph 3.4.1.

(Copies of drawings are available from the U.S. Army Natick Research, Development and Engineering Command, AMSRD-NSC-CF-S, Natick, MA 01760-5018).

6.6 Government-loaned property. The Government will make the following equipment available for loan.

6.6.1 Equipment required for first article and conformance inspection.

Components	National Stock Number	Quantity
Modern Burner Unit (MBU)	7310-01-507-9310	3
Burner Rack	7360-01-250-3645	1
Base Rack	7360-01-250-3652	1
Tray Pack	8920-01-173-1940	20

6.6.2 Equipment required for interoperability and interchangeability demonstrations. Components listed in table II and table III may be provided from current government supply for use as baseline components.

6.8 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided the material meets or exceeds all specified requirements and promotes economically advantageous life cycle costs.

6.9 Subject term (key word) listing.

Enhanced KCLFF
Food heating and serving equipment
Standard KCLFF
Tray pack and A/B rations

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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